

1. Show how a bottom up parser with the given grammar rules will process the following input string. (digit - (id)) (12)

$E \rightarrow E + T \mid E - T \mid E * T \mid E / T \mid T$

$T \rightarrow F \mid C$

$F \rightarrow (E) \mid \text{digit} \mid \text{id}$

$C \rightarrow \text{digit} - \text{id}$

Stack	Input Buffer	Action
\$	(digit - (id)) \$	Shift
\$ (	digit - (id)) \$	Shift
\$ (digit	-(id)) \$	Reduce by $F \rightarrow \text{digit}$
\$ (F	-(id)) \$	Reduce by $T \rightarrow F$
\$ (T	-(id)) \$	Reduce by $E \rightarrow T$
\$ (E	-(id)) \$	Shift
\$ (E -	(id)) \$	Shift
\$ (E - (	id)) \$	Shift
\$ (E - (id	) \$	Reduce by $F \rightarrow \text{id}$
\$ (E - (F	) \$	Reduce by $T \rightarrow F$
\$ (E - (T	) \$	Reduce by $E \rightarrow T$
\$ (E - (E	) \$	Shift
\$ (E - (E)	) \$	Reduce by $F \rightarrow (E)$
\$ (E - F	) \$	Reduce by $T \rightarrow F$
\$ (E - T	) \$	Reduce by $E \rightarrow E - T$
\$ (E	) \$	Shift
\$ (E)	\$	Reduce by $F \rightarrow (E)$
\$ F	\$	Reduce by $T \rightarrow F$
\$ T	\$	Reduce by $E \rightarrow T$
\$ E	\$	Accept